

UNDERGRADUATE RESEARCH PROPOSAL

Venetoclax as a Dual-Action Agent: Apoptosis and Innate Immune Activation via VDAC1 Gate-Opening

Temporal Decoupling of mtDNA Release and Cell Death in AML Cell Lines

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Relevant preprint: github.com/templetwo/vdac-pharmacology-atlas

The Question

Venetoclax (ABT-199) is an FDA-approved drug for acute myeloid leukemia (AML). The standard explanation for how it works: it inhibits BCL-2/Bcl-xL, releasing the apoptotic brake, and the cancer cell dies. This proposal tests a different — and potentially additional — mechanism.

Hypothesis: Venetoclax acts through two temporally distinct mechanisms. First, it displaces Bcl-xL from the VDAC1 channel on the outer mitochondrial membrane, triggering VDAC1 oligomerization and mtDNA release into the cytoplasm. The innate immune sensor cGAS–STING detects this mtDNA and fires an alarm signal. Second — and only afterward — canonical apoptosis (caspase activation, cell death) follows.

If this temporal decoupling is real, venetoclax is not just an apoptosis drug. It is a dual-action agent: an innate immune activator first, a cancer cell killer second. That sequence is unreported and would reframe how the drug is understood — and how it might be combined with checkpoint inhibitors.

Why It Matters

The cGAS–STING pathway is the same pathway that checkpoint inhibitors (anti-PD-1, anti-PD-L1) need to function. If venetoclax activates cGAS–STING before killing cancer cells, then venetoclax + checkpoint inhibitor combinations might have synergistic potential in AML — the innate alarm recruits T cells; the checkpoint inhibitor amplifies them.

This connects to published work showing that mtDNA release activates cGAS–STING (Rongvaux et al., *Cell* 2014; McArthur et al., *Science* 2018) and to the VDAC1–Bcl-xL structural interaction (Shimizu et al., *Nature Cell Biology* 1999; Arbel et al., *EMBO Journal* 2012).

What We Would Measure

The experiment has one central design: treat AML cells with venetoclax and measure the sequence of events — does innate signaling fire before or after cell death?

Timeline: venetoclax treatment at $t = 0$, collect cells at $t = 2\text{h}, 4\text{h}, 8\text{h}, 12\text{h}, 24\text{h}$.

Two parallel treatment arms run simultaneously:

Arm A — Venetoclax alone (standard condition)

Arm B — Venetoclax + z-VAD-FMK (pan-caspase inhibitor, $50\text{ }\mu\text{M}$)

z-VAD-FMK blocks caspase-3 and caspase-7, which — once activated — physically cleave cGAS and IRF3, destroying the innate immune sensor (Rongvaux et al. 2014). Without this inhibitor, even if the gate opens and the alarm fires, dying caspases may degrade the signal before we can measure it. Adding Arm B gives a clean window: the gate can open, mtDNA can flood the cytosol, and the immune signal has nothing suppressing it. If the hypothesis is correct, Arm B should show a dramatically amplified and earlier IFN- β signal than Arm A.

At each time point, three measurements in both arms:

Table 1. Primary measurements at each time point.

Measurement	What It Tells Us	Method	Equipment
Cytosolic mtDNA	Is mtDNA leaking from mitochondria?	Cell fractionation + qPCR	qPCR machine
IFN-β secretion	Is cGAS-STING alarm firing?	ELISA on cell supernatant	Plate reader
Caspase-3 activity	Is the cell dying?	Colorimetric assay	Plate reader

Predicted Outcomes

Positive result: In Arm A, mtDNA and IFN- β rise at 2–4h, caspase-3 follows at 8–12h. In Arm B, IFN- β signal is larger and earlier with caspase-3 suppressed.

Negative result: caspase-3 fires simultaneously with or before mtDNA/IFN- β in both arms. Standard apoptosis explanation is complete.

Either result advances knowledge.

Cell Line

THP-1 (ATCC TIB-202) — human AML monocytic cell line. Suspension culture. Well-characterized for cGAS-STING signaling studies. Intact STING pathway. Primary cell line for this experiment.

Backup: U937 (ATCC CRL-1593.2) — human myeloid leukemia cell line, suspension culture, intact STING pathway. If THP-1 is unavailable, U937 is the appropriate backup.

Important: HL-60 (sometimes listed as an AML alternative) is STING-deficient and must not be used as a backup for this experiment. If HL-60 were substituted and IFN- β was undetectable, the result would be uninterpretable — the sensor would be absent, not the signal.

Controls

- **Vehicle control:** DMSO at matched concentration (no venetoclax)
- **Positive apoptosis control:** Staurosporine (known apoptosis inducer)
- **z-VAD-FMK alone:** Confirms the inhibitor has no baseline effect on IFN- β
- **VDAC1-KO control** (optional, if CRISPR available): should abolish early mtDNA release if the gate-opening mechanism is specific

Equipment Required

Item	Available at DVU?
BSL-2 cell culture hood + incubator	Confirm with lab
qPCR machine	Confirm with lab
Plate reader (absorbance + fluorescence)	Confirm with lab
Centrifuge (for cell fractionation)	Likely yes
Standard micropipettes, tubes, consumables	Yes

Reagents and Estimated Cost

Reagent	Source	Est. Cost
Venetoclax (ABT-199), research grade	Selleck Chemicals (S8048)	~\$150
THP-1 cell line	ATCC or lab gift	~\$300 / free
Cytoplasmic/mitochondrial fractionation kit	Abcam ab65320 or Thermo 89874	~\$200
Human IFN- β ELISA kit	Invivogen or R&D Systems	~\$350
Caspase-3 colorimetric assay kit	Abcam ab39401	~\$150
z-VAD-FMK (pan-caspase inhibitor)	Selleck Chemicals (S7023)	~\$100
qPCR primers (Cyt b / D-loop; GAPDH)	IDT or Sigma	~\$100

Reagent	Source	Est. Cost
Culture media (RPMI + FBS), consumables	Lab stock	~\$100–200
Total		~\$1,400–1,700

Timeline

Week	Activity
1–2	Acquire cells, expand culture, confirm viability
3	Dose-response curve: find venetoclax IC50 in THP-1
4–5	Time-course experiment: 5 time points × 3 measurements
6	Repeat experiment (biological replicate)
7–8	Data analysis, figures, interpretation

Total: ~8 weeks from cell arrival to data.

What Success Looks Like

A positive result — mtDNA in cytosol and IFN- β secretion preceding caspase activation — would be a novel mechanistic finding for venetoclax. It would:

1. Support submission of a short communication to a peer-reviewed journal (e.g., Cancer Letters, Biochemical and Biophysical Research Communications)
2. Provide direct experimental evidence for the gate-jamming framework developed in the accompanying preprint
3. Support a grant application (undergraduate research grant, NIH AREA, or institutional funds) for a larger follow-up study
4. Be a concrete finding to bring to a collaborating AML lab or clinical center

A negative result is equally publishable as a mechanistic constraint.

Anticipated Confound: TREX1

THP-1 cells express TREX1 — a cytosolic DNA exonuclease that degrades single-stranded DNA fragments, including mtDNA. This is directly relevant: our transcriptomic analysis (S2, COADREAD MSS cohort) showed that TREX1 expression rises in lockstep with high gate-jamming scores ($\rho = +0.315$, $p_{\text{bonf}} = 7 \times 10^{-5}$), suggesting that tumors deploy TREX1 as a secondary guard to degrade any mtDNA that leaks past VDAC1 gate-jamming.

In this experiment, TREX1 could produce the following pattern: cytosolic mtDNA appears briefly at 2h (gate opens, mtDNA leaks), then disappears by 4h (TREX1 degrades it), and IFN- β rises only weakly or later than predicted.

If this pattern appears, it is not a failed experiment. It is the S2 discovery manifesting in vitro — the same belt-and-suspenders evasion identified computationally now visible at the cellular level. The discussion section should address this pattern explicitly if observed, and flag TREX1 inhibition as the logical next condition to test.

Literature Basis

Rongvaux A et al. Apoptotic caspases prevent the induction of type I interferons by mitochondrial DNA. *Cell* 2014;159:1563–1577.

McArthur K et al. BAK/BAX macropores facilitate mitochondrial herniation and mtDNA efflux during apoptosis. *Science* 2018;359:eaao6047.

Shimizu S et al. Bcl-2 family proteins regulate the release of apoptogenic cytochrome c by the mitochondrial channel VDAC. *Nature* 1999;399:483–487.

Arbel N et al. VDAC1 as a mediator of the anti-apoptotic activity of Bcl-xL. *EMBO J* 2012;31:1980–1992.

DiNardo CD et al. Venetoclax combined with decitabine or azacitidine in treatment-naïve AML. *NEJM* 2020;383:617–629.